

Manuscript outline: IMRaD plus Threats to Validity and Data Availability

Target venue: ICSE 2027 Software Engineering Education and Training (SEET), Research Paper category

Working title: When Planning Becomes Evidence: An Evidence-Constrained Study of Generative-AI-Supported Software Engineering Education

Submission budget: 10 pages including the main paper, plus up to 2 pages of references under the current call. Recheck the call before submission. The outline reserves space for a short Data Availability statement after the Conclusion and treats the four narrative acts as an argument map rather than as four empirical stages.

Paper-wide evidence rule

The paper distinguishes three statement levels:

- **Observed:** directly reported by a persisted artifact and numerically verified.
- **Interpretive:** a bounded explanation consistent with the reports, structured sources, and literature.
- **Proposed:** a methodological implication or future hypothesis, especially SDD/BDUF; never a demonstrated intervention effect.

The central boundary appears early and repeatedly: team-semester $n = 14$, five tested primary analyses, no significant primary result, and a **conditional-go** verdict. Cross-evidence is secondary exploratory triangulation and cannot replace that verdict.

Research questions

- **RQ1:** What planning, churn, technical-degradation, and temporal patterns are observable in the team-semester evidence?
- **RQ2:** What evidence is available, unavailable, or too sparse to characterize AI-related author concentration and coordination friction?
- **RQ3:** Which secondary cross-evidence patterns are worth carrying forward as bounded hypotheses, and how robust are they to multiplicity, exclusions, and sensitivity checks?
- **RQ4:** What implications for software engineering education and training can be proposed without claiming that SDD/BDUF has been experimentally validated?

Page and evidence budget

paper component	target space	controlling evidence
Introduction	0.75 page	CLAIM-01; agile_manifesto_2001, williams_2024_ai_software_engineering_education

paper component	target space	controlling evidence
Background and Related Work	1.25 pages	cunningham_1992_technical_debt , li_2015_technical_debt_mapping , brooks_1975_mythical_man_month , demarco_lister_1987_peopleware , chen_2021_evaluating_code_llms , perez_2022_planning_and_rework
Method	1.5 pages	CLAIM-02 through CLAIM-07; data book and source manifests
Results	2.5 pages	CLAIM-02 through CLAIM-06; selected figures and primary statistical rows
Discussion	1.5 pages	CLAIM-01, CLAIM-06, CLAIM-07; literature map; Acts 1-4
Threats to Validity	1.0 page	threats_to_validity.md ; all claim threat notes
Conclusion and Data Availability	0.5 page	all retained claims; privacy and artifact-lineage record
References	up to 2 pages	references.bib

1. Introduction

1.1 Educational and generative-AI context

- **Purpose:** establish why software engineering education must examine planning, coordination, and evidence when AI-assisted programming changes the cost and speed of code production.
- **Claim level:** interpretive/contextual, not measured by this dataset.
- **Claims:** CLAIM-01.
- **Evidence:** [artifact report for Act 1](#); [statistical dataset manifest](#).
- **References:** [agile_manifesto_2001](#);
[williams_2024_ai_software_engineering_education](#);
[chen_2021_evaluating_code_llms](#).
- **Allowed wording:** AI-supported course settings make planning and coordination an important educational design question.
- **Avoid:** claiming that AI necessarily causes planning debt or that the present cohort proves a general educational law.

1.2 Problem statement and motivation

- **Purpose:** motivate a study of whether observable planning signals align with later churn, degradation, rework, or coordination signals.
- **Claim level:** interpretive, with the empirical question deliberately narrower than the thesis.

- **Claims:** CLAIM-01, CLAIM-02, CLAIM-03.
- **Evidence:** [Phase 3 data book](#); [argument matrix](#).
- **References:** [cunningham_1992_technical_debt](#); [li_2015_technical_debt_mapping](#); [perez_2022_planning_and_rework](#).
- **Required qualification:** planning intensity is a proxy for artifact volume, not planning quality.

1.3 Qualified hypothesis and research questions

- **Purpose:** state the qualified planning-debt hypothesis and RQ1-RQ4.
- **Claim level:** proposed hypothesis; not a confirmed result.
- **Claims:** CLAIM-01 through CLAIM-07 as the control set.
- **Evidence:** [claim-evidence matrix](#).
- **References:** [cunningham_1992_technical_debt](#); [li_2015_technical_debt_mapping](#).
- **Boundary sentence:** the study tests whether the persisted evidence supports bounded descriptive and exploratory claims; it does not test SDD/BDUF superiority.

1.4 Contributions

Frame three contributions:

1. an empirical characterization of planning, churn, and human-factor signals in an AI-supported SE education cohort;
 2. an auditable evidence organization linking reports, structured artifacts, figures, and caveats;
 3. methodological implications for teaching specification and coordination under AI assistance.
- **Claim level:** observed contribution plus proposed implication.
 - **Claims:** CLAIM-02 through CLAIM-07.
 - **Evidence:** [phase3 evidence inventory](#); [threats record](#).
 - **References:** [williams_2024_ai_software_engineering_education](#); [demarco_lister_1987_peopleware](#).

2. Background and Related Work

2.1 Agile, planning, and specification

- **Purpose:** distinguish adaptive iteration from the question of how much explicit specification is needed before implementation in AI-supported education.
- **Claim level:** theoretical/contextual.
- **Claims:** CLAIM-01; no new empirical claim.
- **References:** [agile_manifesto_2001](#); [brooks_1975_mythical_man_month](#).
- **Evidence link:** [literature map](#).
- **Caution:** SDD/BDUF is introduced as a proposed response, not as an empirically established alternative.

2.2 Planning debt, technical debt, and rework

- **Purpose:** define planning debt as a qualified interpretive construct and relate it to technical debt and rework without claiming equivalence.
- **Claim level:** theoretical.

- **Claims:** CLAIM-02, CLAIM-03, CLAIM-07.
- **References:** [cunningham_1992_technical_debt](#); [li_2015_technical_debt_mapping](#); [perez_2022_planning_and_rework](#).
- **Evidence link:** [data book](#), planning and technical-degradation entries.

2.3 AI-assisted software engineering education

- **Purpose:** position AI author concentration as a measurable but incomplete proxy and identify the educational need for process literacy.
- **Claim level:** contextual and interpretive.
- **Claims:** CLAIM-04.
- **References:** [chen_2021_evaluating_code_llms](#); [williams_2024_ai_software_engineering_education](#).
- **Evidence link:** [argument matrix](#), CLAIM-04.

2.4 Coordination, teamwork, and software engineering skills

- **Purpose:** explain why coordination is theoretically important while acknowledging that the present transcript evidence is sparse and partly unavailable.
- **Claim level:** theoretical, qualified by unavailable evidence.
- **Claims:** CLAIM-05.
- **References:** [brooks_1975_mythical_man_month](#); [demarco_lister_1987_peopeware](#); [miller_2023_coordination_ai_teams](#).
- **Evidence link:** [threats to validity](#), transcript and construct-validity sections.

2.5 SDD/BDUF as the proposed response

- **Purpose:** define specification-driven or front-loaded design as a methodological proposal for future evaluation.
- **Claim level:** proposed only.
- **Claims:** CLAIM-01 and CLAIM-06 provide motivation; no claim asserts superiority.
- **References:** [agile_manifesto_2001](#); [cunningham_1992_technical_debt](#); [brooks_1975_mythical_man_month](#).
- **Evidence link:** [claim matrix](#), especially the conditional-go and exploratory-support cautions.

3. Method

3.1 Educational context, participants, and permitted disclosure

- **Purpose:** describe the course/cohort and team-semester unit without exposing participant, repository, or institution-identifying information.
- **Claim level:** observed methodological description.
- **Claims:** CLAIM-01 context; CLAIM-02 through CLAIM-05 for unit and availability boundaries.
- **Evidence:** [statistical dataset manifest](#); [data book](#).
- **Privacy rule:** exclude private analytical Parquets, raw prompts, raw transcripts, student identifiers, repository identities, and identifying URLs.

3.2 Data sources and evidence hierarchy

- **Purpose:** describe Phase 2.5 reports as interpretive authority, cross-evidence as secondary exploration, and Parquet/CSV/JSON/figure artifacts as verification layers.
- **Claim level:** observed process/method.
- **Evidence:** [phase3 evidence inventory](#); [data book](#).
- **Required statement:** no Phase 2 artifact was regenerated or overwritten for Phase 3.

3.3 Units of analysis and temporal cuts

- **Purpose:** make team-semester, team-semester-cut, file-category, student-response, and transcript-session units explicit.
- **Claim level:** observed methodological description.
- **Claims:** CLAIM-02 through CLAIM-07.
- **Evidence:** [data book](#); [threats record](#).
- **Table candidate:** a compact unit-of-analysis and denominator table, sourced from the data book.

3.4 Metrics and operationalization

- **Purpose:** define PI, CC, AI, DeltaDT, and IE as proxies with their transformations, missingness, and exclusions.
- **Claim level:** observed method plus construct qualification.
- **Claims:** CLAIM-02 through CLAIM-05 and CLAIM-07.
- **Evidence:** [argument matrix](#); [data book](#).
- **References:** [li_2015_technical_debt_mapping](#); [perez_2022_planning_and_rework](#).
- **Required caveat:** PI is not planning quality; CC is not inherently waste; AI is not fully observed tool use; DeltaDT is not all technical debt; IE is not a complete measure of coordination.

3.5 Analysis procedure and missingness handling

- **Purpose:** describe primary correlations, the five-analysis boundary, complete-case handling, unavailable zero-variance analysis, and the exploratory cross-evidence layer.
- **Claim level:** observed method.
- **Claims:** CLAIM-02 through CLAIM-07.
- **Evidence:** [correlation results](#); [statistical manifest](#); [threats record](#).
- **Required statement:** cross-evidence counts are inventory facts, not independent confirmatory tests.

3.6 Reproducibility, privacy, and artifact lineage

- **Purpose:** explain report-first reading, numeric verification, sidecars/manifests, anonymization, and the data-availability boundary.
- **Claim level:** observed process/method.
- **Evidence:** [phase3 evidence inventory](#); [data book](#).
- **Privacy check:** paper-facing outputs contain no raw PII or private evidence fields.

4. Results

4.1 Descriptive dataset profile

- **Purpose:** report the team-semester scope and artifact coverage before interpreting results.
- **Claim level:** observed.

- **Claims:** CLAIM-02 through CLAIM-05.
- **Evidence:** [data book](#); [phase3 inventory](#).
- **Table candidate:** [Table 1](#), compact artifact/unit/row-count/availability profile.
- **Required wording:** identify missing and unavailable measures rather than silently excluding them.

4.2 Report-first evidence map

- **Purpose:** state what the Phase 2.5 artifact and consolidated reports say before presenting cross-evidence.
- **Claim level:** observed report interpretation, reconciled against structured sources.
- **Claims:** CLAIM-02, CLAIM-03, CLAIM-05, CLAIM-07.
- **Evidence:** [consolidated audit](#); [argument matrix](#).
- **Table candidate:** [Table 2](#), claim/status/n/verification path/allowed wording.

4.3 Planning debt and code churn

- **Purpose:** answer the primary part of RQ1: whether initial planning intensity is statistically associated with later code churn or degradation.
- **Claim level:** observed null result.
- **Claims:** CLAIM-02 and CLAIM-03.
- **Evidence:** [correlation results](#); [planning report](#); [technical-degradation report](#).
- **Figures:** [pi_vs_cc](#) as supplementary/null-descriptive; [temporal_escalation_panel](#) as the main descriptive candidate.
- **References:** [cunningham_1992_technical_debt](#); [li_2015_technical_debt_mapping](#).
- **Caption rule:** every caption states team-semester unit and [n](#).

4.4 Integration and human-factor evidence

- **Purpose:** answer RQ2 without overstating sparse AI or transcript measures.
- **Claim level:** observed, qualified, or unavailable.
- **Claims:** CLAIM-04 and CLAIM-05.
- **Evidence:** [integration-friction report](#); [cut-context report](#); [correlation results](#).
- **Figure:** [author_pressure_vs_churn](#) as an exploratory candidate, with [n = 14](#) and the relevant complete-case limitation.
- **Required statement:** transcript coordination friction is unavailable for substantive inference where the primary measure has zero variance.

4.5 Secondary cross-evidence candidates

- **Purpose:** answer RQ3 with exploratory triangulation while preserving the Phase 2.5 verdict.
- **Claim level:** interpretive/secondary exploratory evidence.
- **Claim:** CLAIM-06.
- **Evidence:** [cross-evidence consolidated report](#); [semester-stratified results](#); [plot book](#).
- **Figures:** [scope_vs_late_instability](#); [author_pressure_vs_churn](#).
- **Qualification visuals:** [leave_one_out_robustness](#); [file_category_churn_by_cut](#).
- **Required statement:** favorable cross-evidence rows are candidate associations, not confirmation; the aggregate [supports](#) label does not override the primary conditional-go result.

4.6 Null, unavailable, and mixed results

- **Purpose:** make non-significant and unavailable evidence visible rather than relegating it to limitations.
- **Claim level:** observed.
- **Claims:** CLAIM-02 through CLAIM-07.
- **Evidence:** [argument matrix](#); [threats record](#).
- **Table candidate:** [Table 3](#), primary and secondary result status with n, p-value/status, and interpretation.

5. Discussion

5.1 Act 1: evolutionary ceiling and educational context

- **Purpose:** interpret the educational motivation without presenting Act 1 as a cross-evidence finding.
- **Claim level:** contextual/interpretive.
- **Claims:** CLAIM-01.
- **Evidence:** [Act 1 report](#); [threats record](#).
- **References:** [williams_2024_ai_software_engineering_education](#); [chen_2021_evaluating_code_llms](#).

5.2 Act 2: planning debt as a qualified pattern

- **Purpose:** reconcile descriptive temporal and churn patterns with null primary correlations.
- **Claim level:** observed null plus bounded interpretation.
- **Claims:** CLAIM-02, CLAIM-03, CLAIM-07.
- **Evidence:** [planning report](#); [technical-degradation report](#); [plot book](#).
- **Figures:** [temporal_escalation_panel](#); [file_category_churn_by_cut](#) only as an exclusion warning.
- **References:** [cunningham_1992_technical_debt](#); [li_2015_technical_debt_mapping](#); [perez_2022_planning_and_rework](#).

5.3 Act 3: human factor and coordination boundary

- **Purpose:** discuss author concentration as measurable but sparse and coordination friction as unavailable for substantive inference.
- **Claim level:** descriptive/interpretive with explicit limitation.
- **Claims:** CLAIM-04 and CLAIM-05.
- **Evidence:** [integration-friction report](#); [cut-context report](#); [threats record](#).
- **Figure:** [author_pressure_vs_churn](#) as candidate-level only.
- **References:** [miller_2023_coordination_ai_teams](#); [demarco_lister_1987_peopeware](#); [brooks_1975_mythical_man_month](#).

5.4 Act 4: value inversion and methodological proposal

- **Purpose:** frame SDD/BDUF as a proposal motivated by the evidence boundary, not as a demonstrated solution.
- **Claim level:** proposed.
- **Claims:** CLAIM-01 and CLAIM-06 motivate the proposal; no claim establishes superiority.
- **Evidence:** [argument matrix](#); [threats record](#).
- **References:** [agile_manifesto_2001](#); [brooks_1975_mythical_man_month](#); [cunningham_1992_technical_debt](#).

- **Required wording:** future evaluation should test whether explicit specification and coordination practices improve outcomes under AI assistance.

5.5 Implications for SE education and training

- **Purpose:** derive actionable teaching implications that do not exceed the evidence.
- **Claim level:** proposed implication.
- **Evidence:** CLAIM-01, CLAIM-04, CLAIM-05, CLAIM-07; [threats record](#).
- **References:** [williams_2024_ai_software_engineering_education](#); [demarco_lister_1987_peopleware](#).
- **Permitted implications:** teach planning artifacts as inspectable reasoning, require explicit coordination checkpoints, and assess AI-supported engineering process rather than generated output alone.
- **Avoid:** claiming these practices have been validated as causal improvements by this study.

6. Threats to Validity

6.1 Statistical and internal validity

- **Content:** sample size, unevenness, null tests, zero variance, observational design, multiple comparisons, mixed units, non-paired transcripts, and visual/statistical discrepancies.
- **Evidence:** [threats-to-validity record](#); [correlation results](#).
- **Claims:** CLAIM-02 through CLAIM-06.

6.2 Construct and external validity

- **Content:** PI, CC, AI, DeltaDT, and IE proxy limitations; educational-to-industry transfer; cohort/task dependence.
- **Evidence:** [threats record](#); [data book](#).
- **Claims:** CLAIM-01, CLAIM-04, CLAIM-05.
- **References:** [li_2015_technical_debt_mapping](#); [miller_2023_coordination_ai_teams](#); [demarco_lister_1987_peopleware](#).

6.3 Selection, privacy, and denominator validity

- **Content:** repository survivorship, privacy-driven release restrictions, file-category exclusions, report/source conflicts, and the four exclusion classes.
- **Evidence:** [exclusion manifest](#); [team metrics exclusions](#); [threats record](#).
- **Claims:** CLAIM-07.

7. Conclusion

7.1 Answer to the research questions

- **RQ1:** the artifacts show descriptive planning/churn and temporal patterns, but the primary team-semester relationships are not statistically supported.
- **RQ2:** AI author concentration is measurable in a sparse subset; transcript coordination friction is unavailable or too sparse for a substantive claim.

- **RQ3:** cross-evidence identifies exploratory candidate patterns, but multiplicity, exclusions, and fragility prevent confirmation.
- **RQ4:** the evidence supports SDD/BDUF as a proposal for educational and methodological evaluation, not as an established superior solution.
- **Claims:** CLAIM-02 through CLAIM-07.
- **Evidence:** [argument matrix](#); [threats record](#).

7.2 Final contribution statement

Close with the paper's actual contribution: an evidence-constrained characterization and an auditable way to organize mixed, null, and exploratory results for software engineering education research. Repeat the conditional-go boundary once, without introducing a new claim.

8. Data Availability

Place this section immediately after the Conclusion, as required by the current plan.

- Release the paper-facing aggregate Parquets, CSVs, JSON manifests, figure data, selected figures, data book, argument matrix, and report references that have passed privacy review.
- Do not release raw prompts, raw transcripts, student identifiers, repository identities, private analytical Parquets, or identifying URLs.
- Explain that reviewers can verify the analysis through persisted aggregate artifacts, sidecars, manifests, report chains, and the reproducibility inventory without receiving restricted raw evidence.
- Link the statement to [phase3 evidence inventory](#), [data book](#), and [threats record](#).
- **Privacy status before submission:** run a final double-anonymous and PII scan across all paper-support outputs, figures, metadata, repository history, and external artifact links.

Traceability checklist before drafting prose

- Every empirical paragraph has one or more claim IDs from [argument-matrix.md](#).
- Every number is verified in a persisted CSV, JSON, Parquet, manifest, or figure record through [data_book.md](#).
- Every selected figure is listed in [selected_figures.json](#) and explained in [plot_book.md](#).
- Every literature claim uses a key from [references.bib](#) and is mapped in [literature-map.md](#) and [reference_matrix.csv](#).
- Every favorable cross-evidence statement carries a secondary-exploratory label and its fragility, multiplicity, and exclusion caveat.
- Acts 1 and 4 remain contextual/proposed; they are not presented as empirically proven by cross-evidence.
- The final draft preserves double anonymity and the Data Availability section after the Conclusion.